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MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 7547

This specification is approved for use by the Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, Tunable, 2.0 Mw (Min.) power output, integral magnet, liquid cooled.

ABSOLUTE RATINGS:

Parameter:	If	epy	tk	VSWR	T(anode)	T(bushing)	Output Pressurization
Unit:	A	kv	sec	----	°C	°C	lb_f/in^2
Maximum:	75	62	---	2.0	100	150	40
Minimum:	--	--	600	----	---	---	30
					Note 4	Note 4	Note 5

ABSOLUTE RATINGS:

Parameter:	ib	Ib	Pi	pi	Du	tpc	trv	Coolant Pressure	Tuner Torque
Unit:	mAdc	mAdc	kw	Kw	-----	µs	µs	lb_f/in^2	in-oz.
Maximum:	105	185	11.5	6200	0.002	6.8	----	200	300
Minimum:	60	---	----	----	-----	2.0	0.95	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See outline drawing
Weight:	220 pounds, approximately
Input Bushing:	See outline drawing
Altitude:	See note 6
RF Choke:	See note 1
Mounting Position:	See note 9

TEST CONDITIONS:

Parameter:	Ef	If	tk	tpc	Du	VSWR
Unit:	V	A	sec	µs	-----	(max)
Tolerance:	1.0	-----	---	0.5	-----	---
Test 1	6.5	55	720	---	-----	---
Test 2	6.5	55 Note 20	600	6.0	.0018	1.1

Test Frequencies
F_1 = 420 MHz
F_2 = 435 MHz
F_3 = 450 MHz

REQUIREMENT OR TEST:

Qualification - Not required

Formerly Type QK516

7547

FSC 5960

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNITS
	Quality conformance inspection, part 1						
4003	Pressurizing	-	45 lb_f/in^2 (min) Output only	---	---	---	----
1155	Coolant pressure test	-	315 lb_f/in^2 (min)	---	---	---	----
1155	Anode coolant flow test	-	Flow = 1.0 gal/min, (min) See note 19	P	---	7.0	lb_f/in^2
1155	Tuner coolant flow test	-	Flow = 0.13 gal/min, (min) See note 19	P	---	7.0	lb_f/in^2
1301	Heater current	1	E_f = 5.5 V to 6.5 V	If	---	55	A
1296	Pulse characteristics	2	See notes 13 and 14	---	---	---	----
1256	Average anode current	2		Ib	175	---	mA
4306	Pulse voltage	2	F = F_1, F_2 and F_3	epy	50	60	kv
4250	Power output	2	F = F_1, F_2 and F_3	Po	3750	---	W
----	Tunable frequency	-	T anode = 50 ±25°C See note 22	ΔF	420	450	MHz
4308	Spectrum measurements	2	F = F_1, F_2 and F_3 VSWR = 1.5 (min) See note 15				
4308	RF bandwidth	2	F = F_1, $F_2$and F_3 VSWR = 1.5 (min) See note 15	BW	---	2.0 tpc	MHz
4308	Side lobe height	2	F = F_1, F_2 and F_3 VSWR = 1.5 (min) See note 15	SL	6	---	db
4315	Stability	2	F = F_1, F_2 and F_3 VSWR = 1.5 (min) See notes 23 and 24	MP	---	1	%
4311	Frequency pushing factor	2	F = F_1, F_2 and F_3 See note 17	ΔF	---	0.13	MHz
	Quality conformance inspection, part 2						
4223	Tuner torque	-	Ta = 25°C (approx) See note 18	Torque	---	130	in-oz.
4310	Frequency pulling factor	2	F = F_1, F_2 and F_3 See note 16	ΔF	---	1.2	MHz
4309	Spectrum measurements	2	Ib = 160, 175, and 195 mAdc F = F_1, F_2 and F_3 VSWR = 1.5 (min) See note 15				

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METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	LIMITS
	Quality conformance inspection, part 3						
	Life tests	-	Group D See note 10	t	1000	---	hrs
	Life test end points:						
----	Load stability	2	VSWR = 1.5 F = F_1, F_2 and F_3	MP	---	2	%
4308	Spectrum measurements	2	VSWR = 1.5 (min) F = F_2				
4308	RF bandwidth	2	VSWR = 1.5:1 (min) F = F_1, F_2 and F_3	BW	---	2.0 tpc	MHz
4308	Side lobe height	2	VSWR = 1.5:1 (min) F = F_1, F_2 and F_3	SL	6	---	db
4306	Pulse voltage	2	F = F_1, F_2 and F_3	epy	50	60	kv
4250	Power output	2	F = F_1, F_2 and F_3	Po	3000	---	W
1042	Shock	-	G = 15, Test Method J	---	---	---	---
1031	Vibration	-	See notes 3 and 21	---	---	---	---
----	Tuner life test	-	See notes 8 and 21	Life	500	---	cycles
----	X-radiation	-	See notes 7 and 21	---	---	---	---
4223	Operating torque	-	T = -54°C to +38°C I_f = 55 A	Torque	---	130	in-oz
1011	Humidity	-		---	---	---	---
1143A	Anode cooling test	- -	See notes 4, 11 and 21 Flow = 0.5 gal/min	 ΔTa ΔPa	 --- ---	 87 1.9	 °C lb_f/in^2
		-	Flow = 0.75 gal/min	ΔTa ΔPa	--- ---	76 4.0	°C lb_f/in^2
		-	Flow = 1.0 gal/min	ΔTa ΔPa	--- ---	70 7.0	°C lb_f/in^2
1143A	Tuner cooling test		See notes 12, 19 and 21				
		-	Flow = 0.1 gal/min	ΔTt ΔPt	--- ---	16 1.6	°C lb_f/in^2
		-	Flow = 0.2 gal/min	ΔTt ΔPt	--- ---	13 13	°C lb_f/in^2
		-	Flow = 0.3 gal/min	ΔTt ΔPt	--- ---	11 36	°C lb_f/in^2
----	Non-oscillating characteristics	-	See notes 2 and 21 F = F_1, F_2 and F_3 F = F_1, F_2 and F_3	QL Nc	145 75	250 ---	---- %
----	Position of minimum	-	425 MHz	---	-3	1	cM
----	RF Phase stability	-	VSWR = 1.5 (min) Ib = 160, 175, and 185 mAdc	Δ ϕ	---	3	deg.

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NOTES:

1. The tube shall contain an internal RF choke to minimize radiation from the area of the high voltage bushing. The calculated choke impedance shall be a minimum of 60 ohms over the frequency range.

2. With the magnetron operating per test condition, measurement of interpulse SWR and position of minimum shall be taken using suitable test equipment. The position of minimum shall be measured from the surface defined by Reference Plane D (see outline drawing), and shall be within the limits specified, where minus (-) denotes the direction looking into the output toward the center of the anode diameter. These values shall be within the limits specified.

3. VIBRATION

   Step 1. The magnetron shall be mounted with the cathode vertical, and with rated heater current applied shall be vibrated in plane X_1 with the frequency continually varied from 5-40 Hz and from 50-500 Hz, at a double amplitude excursion of .010 or 2G, whichever is greater. The rate of frequency change shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. A total of four cycles will be performed and the frequencies of all major resonances recorded.

   Step 2. The magnetron shall be vibrated for 30 minutes at each of the major resonances noted in Step 1.

   Step 3. Steps number 1 and number 2 shall be repeated with the tube vibrated in two other mutually perpendicular planes.

4. The temperature is to be measured at the point indicated on the outline drawing.

5. The gas used in pressurization shall provide insulating properties at least equal to that of clean dry air at the pressure indicated.

6. With the tube installed as specified it is capable of satisfactory operation at altitudes up to 20,000 ft. The tube nonoperating can withstand transportation or storage at altitudes up to 75,000 feet.

7. The tube shall be covered as required by shielding, excluding the input and output connections which will reduce X-radiation to a safe level. U.S. Department of Commerce Handbook 60, "X-ray Protection", shall be used as a guide for radiation considerations.

8. The Tuner Life Test shall be conducted under the following conditions:

   a. Filament Current 55 amperes
   b. Anode Temperature 50-60°C
   c. Tuner Temperature 60°C approximately
   d. Drive mechanism set for 18 revolutions, in each direction, at 30 rpm.

9. The tube may be mounted with the cathode vertical within 45° in any direction provided the total operating time is limited to 15 minutes; for any continuous operation the cathode will be vertical within 15 degrees.

10. Power input to the tube during life test shall be cycled according to the following schedule:

	Condition	Time	If	Ib
a.	Preheat	600 sec (max)	55A	0
b.	Test 2	3.5 hr.		
c.	Tube off	20 min.		

The accumulation of the "on time" shall be the only time considered in determining compliance with the minimum time value.

Frequency of operation during life test shall be reset once every twenty-four hours, and shall proceed according to the following schedule:

406 to 425 to 450 to 406 to 425 MHz, etc.

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NOTES: -Continued

11. ΔTa is the temperature rise of the anode above that of the coolant temperature at the inlet. ΔPa is the pressure drop across the anode coolant jacket.

12. ΔTt is the temperature rise of the coolant between the inlet and outlet of the tuner cooling system. ΔPt is the pressure drop across the tuner cooling system.

13. This test need be conducted only under one set of conditions within the limits stated for the oscillation specified.

14. The pulse characteristics for Test 2 operation shall be as follows:

	Min.	Max.	
trv:	---	0.95	μsec (measured 20 to 85%)
tfc:	---	2.5	μsec (measured 0 to 85%)
tfv:	---	6.0	μsec (measured 0 to 85%)

No spike or ripple shall exceed ±11% of the average peak value of voltage or current. Inverse voltage shall not exceed 60% of the forward voltage. The negative amplitude of any post pulse voltage oscillation shall be kept to zero.

15. Stability will not be measured under this test. The RF bandwidth shall be within the limits specified when a VSWR of 1.5:1 is introduced in the load at a distance of approximately 1 meter from the magnetron coupling flange, the phase being adjusted for the widest spectrum.

16. The pulling measurement shall be made in such a manner that thermal effects do not introduce appreciable errors.

17. The anode current shall be varied between 160 mAdc and 185 mAdc. Thermal effects shall be minimized. The ΔF measurement is the difference between the frequency extremes observed.

18. The tuner drive mechanism shall not be "set" against the mechanical stops.

19. Water shall be the coolant used for this test. There shall be no evidence of leaks.

20. The heater surge current shall not exceed 135 amperes.

21. If these tests have been performed previously by the manufacturer and the materials, the manufacturing processes, the test requirements are the same as when the tests were performed, these tests need not be performed.

22. The tuner mechanism shall be provided with a sealed stop device restricting the lower frequency limit to 422.2 ±2 MHz.

23. No more than the specified percentage of pulses shall result in RF output pulses each having less than 70% of the energy content of a normal pulse. The test will be conducted for a period not to exceed 15 minutes, and satisfactory operation must be obtained for the last five consecutive minutes of the test period. A missing pulse is defined as a pulse, the energy of which, within a ±1% frequency band at the mean normal test frequency, is less than 70% of the energy of a normal pulse. The test shall be conducted at the VSWR phase of greatest instability.

24. The stability test at F_1 shall be the first test performed after holding period.

Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

Review activity:
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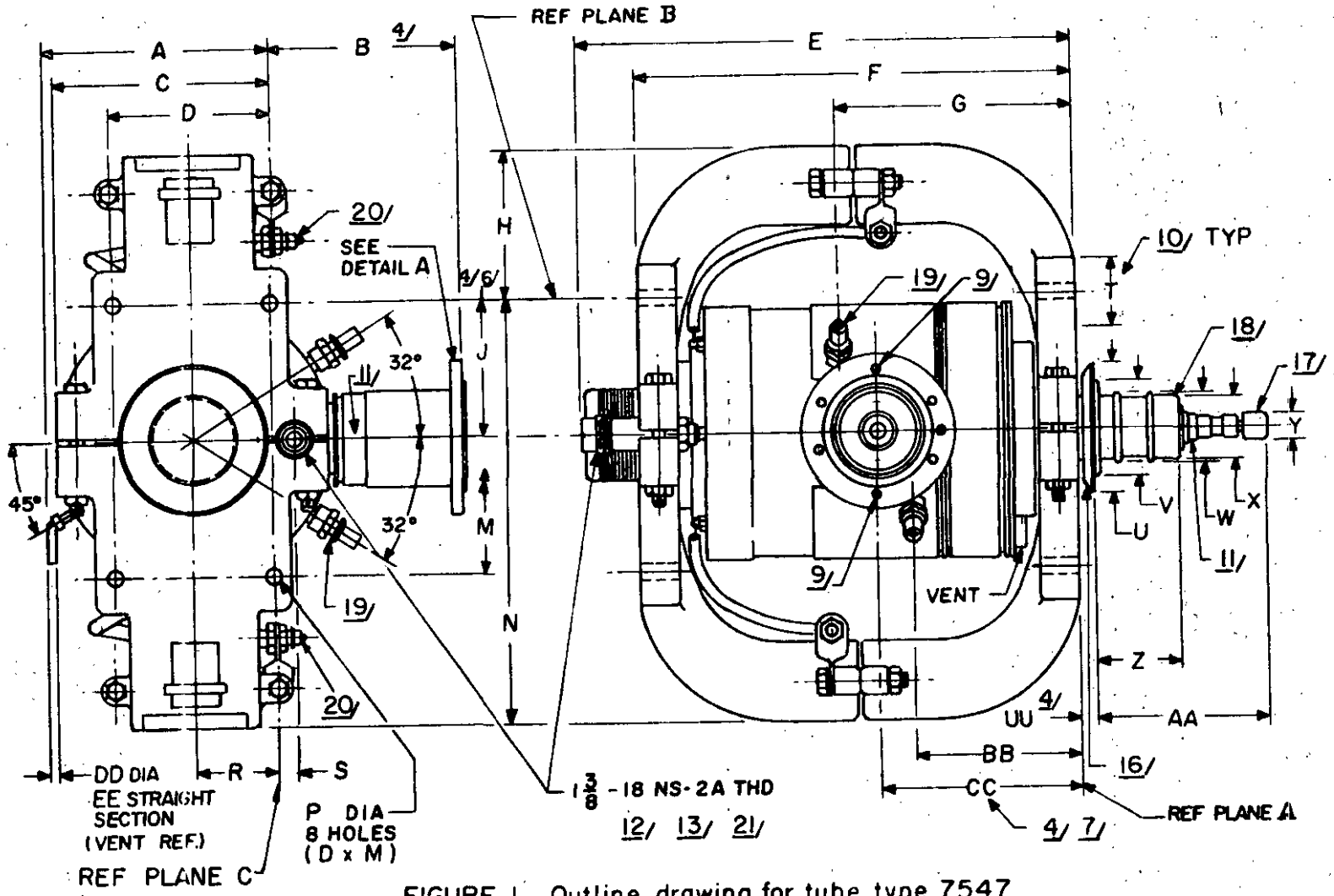
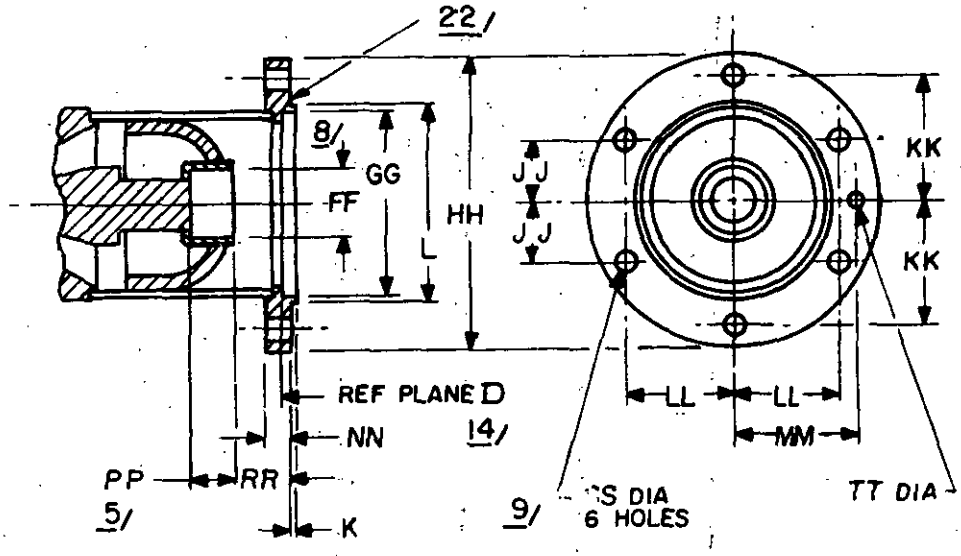


FIGURE 1 Outline drawing for tube type 7547

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DIMENSIONS

SYM	INCHES		MM	
	MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 1				
B	6.469	6.843	164.31	173.81
C		7.437		188.90
E		16.750		425.45
H		5.437		138.10
J	4.313	4.687	109.55	119.05
M	8.950	9.050	277.33	229.87
N		14.437		366.70
P	.612	.628	15.54	15.95
T	2.000		50.80	
Z	2.721	2.971	69.11	75.46
AA	5.491	5.875	139.47	149.23
CC	6.390	6.770	162.31	171.96
UU	.486	.548	12.34	13.92
QUALITY CONFORMANCE INSPECTION, PART 2				
D	5.465	5.535	138.81	140.59
F		14.625		371.48
R	2.625	2.875	66.68	73.03
S	.625	.875	15.88	22.23
U	4.485	4.515	113.92	114.68
X	2.346	2.370	59.59	60.20
Y	.868	.878	22.05	22.30
FF	1.227	1.247	31.17	31.67

SYM	INCHES		MM	
	MIN	MAX	MIN	MAX
JJ	1.084	1.104	27.53	28.04
KK	2.177	2.197	55.30	55.80
LL	1.884	1.904	47.85	48.36
PP	.750		19.05	
MM	2.177	2.197	55.30	55.80
RR	.860	1.000	21.84	25.40
SS	.398	.414	10.11	10.52
TT	.248	.256	6.30	6.50
QUALITY CONFORMANCE INSPECTION, PART 3				
A		8.000		203.20
K	.014	.018	.36	.46
L	3.323	3.333	84.40	84.66
W		2.760		70.10
EE	.750		19.05	
GG	3.198	3.208	81.23	81.48
NOMINAL DIMENSIONS				
G	7.830		198.88	
V	3.485		88.52	
BB	5.330		135.38	
DD	.250		6.35	
HH	5.156		130.96	
NN	.437		11.10	

FIGURE 1 Outline drawing for tube type 7547 (continued)

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NOTES:

1. Reference plane "A" is defined as a plane passing along the face of the mounting surface as shown.
2. Reference plane "B" is defined as a plane perpendicular to plane "A" and passing through the center of holes at plane "A" as shown.
3. Reference plane "C" is defined as a plane perpendicular to planes "A" and "B" passing through the centerline of output side mounting holes as shown.
4. Includes angular as well as lateral deviation.
5. Applies to straight portion of inner conductor wall.
6. Parts on this centerline may vary from the true location by .125.
7. Applies to center of diameter "GG."
8. Centerline of inner conductor shall be concentric with centerline of outer conductor within .025.
9. Centerline passing through these holes must be parallel to reference plane "A" within .050 T.I.R.
10. These surfaces must be coplanar within .015.
11. Temperature measuring points.
12. Pitch diameter must accept class 1 "go" gage only.
13. Major diameter must not be less than 1.362.
14. Reference plane "D" is defined as a plane along the surface of the output flange as indicated.
15. Epoxy base paint having equivalent properties of Durachem basing enamel to be used.
16. Oil bushing flange.
17. Heater connector.
18. Common heater-cathode connector.
19. Anode coolant connector (Wiggins 6005R64W6) mates with E. B. Wiggins self sealing type socket 6000-8 series.
20. Tuner coolant connector (Wiggins 600R62W6) mates with E. B. Wiggins self sealing type socket 6000-6 series.
21. Tuner drive coupling to mate with S. S. White combination = 3467Y-0G or equivalent.
22. Output coupling flange mates with Andrew Corp. Type 562 3 1/8 LHF transmission line or equivalent.
23. Metric equivalents (to the nearest .01 mm) are given for general information and are based upon 25.4 mm = 1 inch.

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